

Power Take-Off Design and Analysis for a Tunable Laboratory-Scale Wave Energy Converter for Powering Blue Economy Applications [★]

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Abstract: This work presents the numerical characterization of the Power Take-Off (PTO) system for a tunable laboratory-scale two-body Point Absorber (PA) Wave Energy Converter (WEC). Three PTO architectures are developed and compared using time-domain simulations in WEC-Sim under regular wave conditions: an ideal Spring-Damper (SD) baseline, a Direct-Drive Linear Generator (DDLG), and an Electrical Generator Equivalent Circuit (EGEC). Under quasi-static electrical assumptions, both electrical systems can be parameterized as a tunable linear damper, whose coefficient is set to match the optimal damping target. The DDLG is tuned through its electrical load resistance, while the EGEC offers an additional tuning parameter through its rack-and-pinion transmission ratio. Our study demonstrates deficiencies in the existing WEC-Sim model, and applies efficiency corrections to obtain experimentally consistent power estimates. The results establish a validated numerical baseline for the forthcoming experimental campaign at the University of New Hampshire (UNH) wave tank, with predicted mechanical power outputs reaching up to 37 [W] and corrected electrical outputs of approximately 26 [W] and 21 [W] for the DDLG and EGEC, respectively, at a wave frequency of $\omega = 4.0$ [rad/s] and wave amplitude of $\eta = 0.16$ [m].

Keywords: Wave Energy Converters (WECs), Power Take-Off(PTO), Powering Blue Economy (PBE), WEC-Sim.

1. INTRODUCTION

Marine Energy (ME) has emerged as a promising contributor to the de-carbonization of the power production sector by harnessing the dense available energy contained in ocean waves, tides, and currents. In the United States (U.S.), the technical potential of these resources is estimated to satisfy up to 50% of the national electricity demand, according to U.S. Department of Energy, Water Power Technologies Office (2025). Despite significant progress in numerical modeling and hydrodynamic optimization, challenges persist in integrating ME systems into the main power grid; therefore, large-scale deployment of WECs remains limited. Hence, focus has been shifted to Powering Blue Economy (PBE) applications defined as energy solutions and technological innovations that support ocean-based activities in remote environments often far from the traditional electrical grid, such as aquaculture or offshore charging stations for autonomous underwater vehicles, LiVecchi et al. (2019).

Wave Energy Converters (WECs) represent a key class of technologies to extract high-density power from ocean wave motion, as described by Falnes and Kurniawan (2020). Among the various WEC architectures, point absorbers (PAs) have proven to be a particularly promising type, owing to their compact footprint, omni-directional wave capture capability, and favorable power-to-mass ratio (PMR), Falnes and Kurniawan (2020). Specifically, two-body PA systems enable an enhanced energy extraction capability through relative motion between a floating body and a submerged reaction structure, known as a heave-plate. This configuration offers additional flexibility for dynamic tuning and controls.

The Reference Model 3 (RM3), developed under the U.S. Department of Energy's (DoE) Reference Model Project (RMP), provides a well established and extensively validated benchmark for two-body PA WECs, see Neary et al. (2014). Originally scaled for grid-scale applications, it has undergone intensive numerical and experimental analysis, resulting in a robust, well-documented dataset. Its open-access hydrodynamic data and widespread use within the research community make it an ideal foundation for investigating scaled, tunable WEC designs. Although the hydrodynamic performance of such systems is well characterized, understood, and easily accessible by open-source

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software such as WEC-Sim Ruehl et al. (2014), the design and implementation of physical representative Power Take-Off (PTO) systems remains a critical challenge.

The PTO system is a critical component of any WEC, as it largely dictates the efficiency, cost, and control of the overall system, So et al. (2015); Zhang et al. (2023). Governing both efficiency and mechanical-to-electrical power conversion, this system cannot be misrepresented in modeling. The most commonly employed modeling approach represents the PTO as an ideal linear Spring-Damper (SD), which provides a tractable analytical framework for WEC characterization but neglects the electromechanical dynamics inherent to real devices, Falnes and Kurniawan (2020). More physically representative alternatives has been studied in the context of two-body PA, including hydraulic PTOs, described in So et al. (2015); Direct Drive Linear Generators (DDLG), following the approach shown in Mueller (2002) and Zhang et al. (2023); and rotary generator systems coupled through mechanical transmissions as presented in Polinder et al. (2005). Bridging the gap between idealized SD models and physically realizable PTO architectures is therefore essential for translating numerical designs into functional prototypes.

This work addresses the challenge of investigating and comparing multiple PTO representations for a tunable laboratory-scale two-body PA WEC. It builds upon the authors' previous study, in which the optimal WEC design, including the optimal damping coefficient, was identified using frequency-domain surrogate modeling based on a Boundary Element Method (BEM). However, detailed PTO modeling requires a transition to the time-domain to capture additional system dynamics. Specifically, three PTO models are modeled, improved, and implemented within the time-domain WEC-Sim simulation framework: an ideal linear spring-damper (SD) model, a direct-drive linear generator (DDLG), and an electrical generator equivalent circuit (EGEC). Each model is parameterized using a common optimal damping coefficient, B_{PTO} , obtained from the prior BEM-based optimization Matamala-Carvajal et al. (2026), enabling a consistent comparison between idealized and physically realizable PTO behaviors. The electrical power outputs from the DDLG, EGEC, and their equivalent SD model are evaluated over a range of wave conditions representative of the University of New Hampshire (UNH) wave tank capabilities. This analysis provides a basis for selecting and sizing PTO systems for future experimental implementation. The WEC-Sim numerical framework developed in this work is publicly available through the project GITHUB repository¹.

2. WAVE ENERGY CONVERTER DESIGN AND HYDRODYNAMIC MODELING

2.1 Tunable Laboratory-Scale Point Absorber Wave Energy Converter

This study focuses on a two-body PA WEC design whose geometry is similar to the well-established RM3 device, in which a floating buoy oscillates relative to a submerged

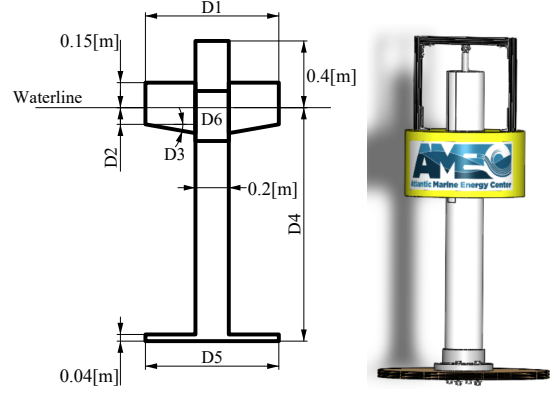


Fig. 1. Left: Schematic of the laboratory-scale two-body point absorber (PA) wave energy converter (WEC), illustrating the six primary design variables ($D1$ – $D6$). Right: Final prototype selected for simulation and experimental testing at the University of New Hampshire (UNH) wave tank.

spar-plate configuration, in response to ocean wave excitation Neary et al. (2014). In this work, a tunable laboratory-scale adaptation of the RM3 geometry is investigated. The scaled model was derived through a BEM-based optimization process described in authors' previous paper Matamala-Carvajal et al. (2026), from which optimal dimensions and linear PTO damping coefficient B_{PTO} were obtained according to the operational constraints of the UNH wave tank, information available at University of New Hampshire (2025).

A simple schematic of our WEC and a 3D model of the final prototype is shown in Fig. 1. The main dimensions of the WEC are a buoy diameter of $D1 = 0.7$ [m], draft $D2 = 0.1$ [m], hull angle $D3 = 0^\circ$, spar draft $D4 = 1.3$ [m], plate diameter $D5 = 1$ [m], and an optimal PTO damping coefficient $B_{PTO} = 540$ [kg/s]. While the physical prototype is currently under construction, the present work focuses on the numerical characterization of this geometry through WEC-Sim time-domain simulations.

2.2 Hydrodynamic Modeling

This section presents the hydrodynamic model of the WEC. The equation of motion governing a two-body PA device is formulated as the sum of all hydrodynamic and mechanical forces acting on the system, as given in Eq. 1, according to Falnes and Kurniawan (2020); Journée and Massie (2001):

$$M \ddot{\xi}(t) = F_{hs}(t) + F_{ext}(t) + F_{rad}(t) + F_v(t) + F_m(t) + F_{nh}(t) + F_{PTO}(t) \quad (1)$$

Each term represents a distinct physical contribution, $F_{hs}(t) = -C\xi(t)$ is the hydrostatic restoring force, proportional to the restoration coefficient of the body C and instantaneous displacement; $F_{ext}(t)$ is the wave excitation force, comprising the Froude-Krylov and diffraction components computed via the BEM solver CAPYTAINE, Ansellin and Dias (2019); $F_{rad}(t) = -A(\omega)\ddot{\xi}(t) - B_h(\omega)\dot{\xi}(t)$ is the radiation force, incorporating the frequency-dependent added mass $A(\omega)$ and radiation damping $B_h(\omega)$, related to the acceleration and velocity, respectively; $F_v(t) = -B_v\dot{\xi}(t)$ is the viscous drag force governed by the viscous

¹ <https://github.com/Energy-Control-and-Optimization-Lab/WEC-Tunable-RM/tree/MECC2026>

damping B_v ; $F_m(t)$ is the mooring restoring force; $F_{nh}(t)$ accounts for nonlinear hydrodynamic effects; and $F_{PTO}(t)$ is the PTO force.

Since the PTO mechanism couples both bodies, it acts with equal magnitude and opposite sign on each. The PTO force on both, the buoy and the spar, depends on the relative displacement $\xi_{rel}(t) = \xi_1(t) - \xi_2(t)$ and relative velocity $\dot{\xi}_{rel}(t) = \dot{\xi}_1(t) - \dot{\xi}_2(t)$ between the two bodies, being governed by a damping coefficient B_{PTO} and a stiffness coefficient K_{PTO} . The PTO force is assembled as shown in Eq. 2, according to Ruehl et al. (2014).

$$\mathbf{F}_{PTO} = \begin{bmatrix} F_{PTO}(t) \\ -F_{PTO}(t) \end{bmatrix} \quad (2)$$

In matrix form, the coupled equations of motion for the two-body system expressed in terms of frequency-dependent hydrodynamic coefficients are shown in Eq. 3, according to Journée and Massie (2001); Falnes and Kurniawan (2020).

$$\begin{aligned} &(\mathbf{A}(\omega) + \mathbf{M})\ddot{\boldsymbol{\xi}}(t) \\ &+ (\mathbf{B}_h(\omega) + \mathbf{B}_v + \mathbf{B}_{PTO})\dot{\boldsymbol{\xi}}(t) \\ &+ (\mathbf{C} + K_{PTO})\boldsymbol{\xi}(t) = \mathbf{F}_e(t) \end{aligned} \quad (3)$$

$\boldsymbol{\xi}(t)$, $\dot{\boldsymbol{\xi}}(t)$, $\ddot{\boldsymbol{\xi}}(t)$ and $\mathbf{F}_e(t)$ are 2×1 vectors, and all the other elements are matrix coefficients of size 2×2 . Defining the total damping matrix as $\mathbf{B}(\omega) = \mathbf{B}_h(\omega) + \mathbf{B}_v + \mathbf{B}_{PTO}$, and neglecting K_{PTO} to have a pure dissipative system, the frequency-domain solution (Response Amplitude Operator, RAO) is obtained as reported in Journée and Massie (2001) in Eq. 4.

$$\boldsymbol{\xi}(\omega) = \left[-\omega^2(\mathbf{M} + \mathbf{A}(\omega)) + i\omega\mathbf{B}(\omega) + \mathbf{C} \right]^{-1} \mathbf{F}_e(\omega) \quad (4)$$

The hydrodynamic coefficients $\mathbf{A}(\omega)$, $\mathbf{B}_h(\omega)$, and the excitation force vector $\mathbf{F}_e(\omega)$, required to evaluate Eqs. 3–4, are computed in the frequency domain using the open-source BEM solver CAPYTAINE, Ancellin and Dias (2019). The resulting hydrodynamic data are subsequently exported via BEMIO and used as input to WEC-Sim, which numerically integrates the equations of motion described above to predict the dynamic response of the device and the power extracted by the PTO system under prescribed wave conditions, Ruehl et al. (2014).

3. POWER TAKE-OFF MODELING

In this paper, two PTO architectures are investigated: a DDLG, in which the relative motion between the two bodies is directly coupled to a permanent-magnet linear generator without intermediate mechanical transmission Zhang et al. (2023); and an EGEC, which represents the electromechanical dynamics of a rotary generator and its mechanical transmission using an equivalent circuit formulation. In general, the PTO systems are modeled through time-domain simulations in WEC-Sim Ruehl et al. (2014), where the PTO is implemented using PTO-Sim So et al. (2015), a modular Simulink library that extends the single-degree-of-freedom (SD) formulation to support physically detailed PTO representations.

3.1 Direct Drive Linear Generator (DDLG)

The DDLG is a PTO architecture in which the relative heave motion between the buoy and the spar is mechan-

ically coupled to the translator of a permanent magnet linear generator, converting the kinetic energy directly into electrical energy without intermediate stages, see Mueller (2002) and Zhang et al. (2023). The electromagnetic braking force produced by the generator acts as the PTO force on the WEC, and its magnitude can be controlled through the electrical resistance load, making the DDLG a naturally tunable system. The total mechanical force exerted by the DDLG on the two-body system has three contributions, shown in Eq. 5 Mueller (2002); Mueller and Baker (2005):

$$F_{PTO}^{DDLG}(t) = F_{em}(t) + F_{mec}(t) + F_{cog}(t) \quad (5)$$

$F_{em}(t) = K_f i(t)$ is the electromagnetic braking force, proportional to the armature current $i(t)$ through the generator force constant K_f [N/A]; $F_{mec}(t)$ is the mechanical friction force due to bearing losses and windage; and $F_{cog}(t)$ is the cogging force arising from the interaction between the permanent magnets and the stator teeth. In the present work, $F_{mec}(t)$ is not considered since its inclusion would require additional parameters that cannot be identified from the available data. $F_{cog}(t)$ is neglected since its amplitude is generally less than 5% of the rated electromagnetic force in well-designed permanent magnet linear generators, as is referred in Mueller and Baker (2005). Then, the DDLG force is reduced just to consider the electromagnetic braking force, as shown in Eq. 6:

$$F_{PTO}^{DDLG}(t) \approx F_{em}(t) = K_f i(t) \quad (6)$$

The armature current $i(t)$ model is shown in Eq. 7, and is governed by the electrical dynamics of the generator circuit. Neglecting winding inductance ($L \approx 0$), which is valid when $\omega L \ll R_s + R_L$, the back-EMF $e(t) = K_e \dot{\xi}_{rel}(t)$ drives a quasi-static current through the series combination of the stator resistance R_s and the load resistance R_L .

$$i(t) = \frac{K_e}{R_s + R_L} \dot{\xi}_{rel}(t) \quad (7)$$

K_e [V·s/m] is the back-EMF constant of the generator. Substituting Eq. 7 into Eq. 6 results in Eq. 8:

$$F_{PTO}^{DDLG}(t) \approx \frac{K_f K_e}{R_s + R_L} \dot{\xi}_{rel}(t) \quad (8)$$

3.2 Electric Generator Equivalent Circuit (EGEC)

Unlike the DDLG, the EGEC model represents a rotary permanent magnet generator coupled to the WEC through a rack-and-pinion transmission, which converts the linear relative motion of the two bodies into shaft rotation, Polinder et al. (2005) and So et al. (2015). The generator is modeled as a lumped-parameter equivalent circuit consisting of a back-EMF source, armature resistance, and winding inductance, which captures the essential electromechanical energy conversion without requiring a detailed electromagnetic design. The linear relative velocity $\dot{\xi}_{rel}(t)$ is converted to rotor angular velocity through the rack-and-pinion of pitch radius r , according to So et al. (2015), as shown in Eq. 9.

$$\dot{\theta}(t) = \frac{1}{r} \dot{\xi}_{rel}(t) \quad (9)$$

This rotation induces a back-EMF $e(t) = K_e \dot{\theta}(t)$, which drives current through the full armature circuit as shown in Eq. 10.

$$L_a \frac{di}{dt} + (R_a + R_{load}) i(t) = \frac{K_e}{r} \dot{\xi}_{rel}(t) \quad (10)$$

L_a is the armature inductance, R_a the winding resistance, and R_{load} an external electrical load resistance. At the wave frequencies of interest ω , the inductive term is negligible when $\omega L_a \ll R_a + R_{load}$; under this quasi-static assumption, Eq. 10 reduces to Eq. 11.

$$i(t) = \frac{K_e}{r(R_a + R_{load})} \dot{\xi}_{rel}(t) \quad (11)$$

The electromagnetic torque $\tau_{em}(t) = K_e i(t)$ is referred back to the linear domain through the transmission, yielding the PTO force, illustrated in Eq. 12.

$$F_{PTO}^{EGEC}(t) = \frac{\tau_{em}(t)}{r} = \frac{K_e^2}{r^2(R_a + R_{load})} \dot{\xi}_{rel}(t) \quad (12)$$

3.3 Spring-Damper (SD) Ideal Model

For simplicity, the PTO is commonly modeled as a linear SD system whose force depends on the relative kinematics of the two bodies Falnes and Kurniawan (2020), as expressed in Eq. 13:

$$F_{PTO}(t) = -B_{PTO} \dot{\xi}_{rel}(t) - K_{PTO} \xi_{rel}(t) \quad (13)$$

In the present work, the spring term is neglected by setting $K_{PTO} = 0$, reducing the PTO to a purely dissipative linear damper characterized by the damping coefficient B_{PTO} . This assumption is consistent with passive damping control strategies commonly employed in point absorber (PA) WECs, Falnes and Kurniawan (2020), where the stiffness contribution is either absent or effectively incorporated into the hydrostatic restoring force.

The SD formulation is adopted for both DDLG and EGEC architectures. For the DDLG, under the stated assumptions in Section 3.1 and following Eq. 8, the system can be represented as an equivalent linear damper with an effective PTO damping coefficient B_{PTO}^{DDLG} , given by Eq. 14. This formulation enables B_{PTO}^{DDLG} to be tuned through the load resistance R_L .

$$B_{PTO}^{DDLG} = \frac{K_f K_e}{R_s + R_L} \quad (14)$$

In the same way, the EGEC is simplified to a purely dissipative linear damper with an effective PTO damping coefficient shown in Eq. 15. B_{PTO}^{EGEC} can be adjusted independently by the load resistance R_{load} or the transmission radius r .

$$B_{PTO}^{EGEC} = \frac{K_e^2}{r^2(R_a + R_{load})} \quad (15)$$

Eqs. (14) and (15) establish a general framework for tuning the DDLG and EGEC PTO parameters based on the optimal damping coefficient obtained from frequency-domain optimization. The surrogate modeling approach yields a single optimal value of B_{PTO} , Matamala-Carvajal et al. (2026), which must then be mapped to the physical parameters of the detailed PTO models. This step is essential for complex WEC systems, where optimization is first performed using a simplified surrogate model, followed by detailed PTO modeling to establish the connection

between the lumped damping coefficient and the physical parameters of each specific PTO architecture.

4. SIMULATION SETUP AND MODEL PARAMETERS

All simulations are conducted in WEC-Sim using the two-body PA model described in Section 2.1, with hydrodynamic data imported from the BEM-code CAPY-TAINE via BEMIO. The simulation time step is set to $\Delta t = 0.00025$ [s] to ensure numerical stability, with a total duration of $t_{end} = 120$ [s]. A ramp function of $t_{ramp} = 30$ [s] is applied at the start of each simulation to gradually introduce the wave forcing and avoid impulsive transient loads, following standard WEC-Sim practice, shown in Ruehl et al. (2014). Both bodies are restricted to the heave degree of freedom (vertical motion). A linear viscous damping coefficient of $B_v = 257.64$ [N·s/m] is applied to the spar in the heave direction, while for the buoy the viscosity is neglected ($B_v = 0$).

Three PTO cases are considered, as described in the previous section. An ideal linear SD PTO is modeled based on Eq. 13. The damping coefficient is set to the optimal value $B_{PTO} = 540$ kg/s, while the spring stiffness is set to $K_{PTO} = 0$. This baseline case serves as a reference for evaluating the performance of both PTO architectures and enables assessment of the hydrodynamic response independently of electromechanical losses or transmission dynamics.

The magnetic circuit of the DDLG model is characterized by a pole pitch $\tau_p = 15.2$ [mm], a magnet outer diameter of 50.8 [mm], and a coil inner diameter of 57.4 [mm], resulting in a radial air gap of $g = 3.3$ [mm]. The winding consists of $N = 560$ turns with a measured armature resistance $R_a = 1.1$ [Ω] and inductance $L_a = 13$ [mH]. The force constant K_f is computed analytically from the peak flux per pole, incorporating a fringing correction factor $k_{fringe} = 0.75$ to account for leakage at the magnet edges. The load resistance R_{load} is then determined by inverting Eq. 14 to match the target damping coefficient $B_{PTO} = 540$ [kg/s].

The EGEC model is parameterized using the specifications of the ADR175-B143 three-phase permanent magnet rotary motor, information available at Akribis Systems Pte Ltd (2024). The per-phase armature resistance and inductance are $R_a = 0.89$ [Ω] and $L_a = 5$ [mH], respectively. The back-EMF constant is $K_e = 0.43$ [V/rpm], and the torque constant is $K_t = 5.0$ [N·m/A_{rms}]. The transmission has a pinion radius $r = 43$ [mm], yielding a gear ratio $N_g = 1/r$. As with DDLG, the load resistance R_{load} is determined by inverting Eq. 15 to match the same target damping coefficient $B_{PTO} = 540$ [kg/s], ensuring a consistent comparison between both PTO architectures.

All three PTO configurations (SD, DDLG, and EGEC) are evaluated under the same set of regular wave conditions, defined to match the operational capabilities of the UNH wave tank University of New Hampshire (2025). Seven wave frequencies spanning are considered, with the maximum achievable wave amplitude at each frequency as follows: (ω [rad/s], η [m]) = (1.6, 0.05), (2.0, 0.07), (2.4, 0.09), (2.8, 0.12), (3.2, 0.14), (3.6, 0.16), and (4.0, 0.16).

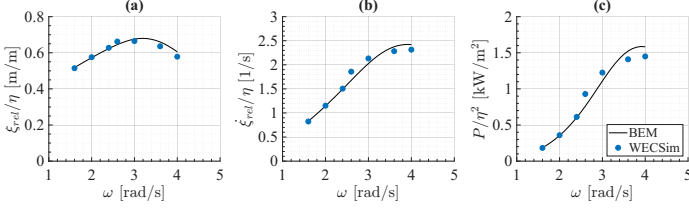


Fig. 2. Comparison between CAPYTAINE frequency-domain analysis (solid line) and WEC-Sim time-domain simulations (blue markers) for the ideal Spring-Damper (SD) PTO, normalized by the incident wave amplitude η . (a) Relative displacement $|\xi_{rel}|/\eta$ [m/m], (b) relative velocity $|\dot{\xi}_{rel}|/\eta$ [1/s], and (c) normalized mean absorbed power P/η^2 [kW/m²].

5. SIMULATION RESULTS AND DISCUSSIONS

This section presents the simulation results from WEC and three PTO configurations. Fig. 2 shows the frequency-domain comparison between the BEM results obtained with CAPYTAINE and the average WEC-Sim time-domain simulations for the ideal SD PTO case. All results are normalized by the incident wave amplitude η to yield amplitude-independent transfer functions. Fig. 2.a represents the relative displacement RAO $|\xi_{rel}|/\eta$ [m/m], Fig. 2.b shows the relative velocity RAO $|\dot{\xi}_{rel}|/\eta$ [1/s], and Fig. 2.c demonstrates the mean absorbed power P/η^2 [W/m²]. In all three figures, the WEC-Sim results (blue markers) show good agreement with the BEM frequency-domain solution (solid black line), with minor deviations at higher frequencies attributable to transient effects in the time-domain integration. This validation confirms that the linear potential flow assumptions underlying the BEM analysis are adequate for the wave conditions and device scales considered, and establishes confidence in the WEC-Sim numerical setup as a reliable tool for PTO performance evaluation. The consistency between both modeling approaches also supports the use of the BEM-derived optimal damping coefficient $B_{PTO} = 540$ [kg/s] as a meaningful design target for the physical PTO systems investigated.

Fig. 3 represents the mean absolute mechanical and electrical power output for the three PTO configurations across the seven wave conditions evaluated, with the corresponding wave amplitude η overlaid on the right axis. For all configurations, absorbed power increases with wave frequency and wave amplitude, reaching a maximum of approximately 37 [W] at $\omega = 4.0$ [rad/s]. The DDLG and EGEC electrical outputs, as computed directly from WEC-Sim, exhibit conversion efficiencies in excess of 95% relative to the mechanical reference. This result is physically unrealistic, as it reflects the absence of copper losses, iron losses, and mechanical friction in the lumped-parameter electrical models implemented in PTO-Sim, So et al. (2015). To provide a more representative estimate of actual electrical output, mechanical-to-electrical conversion efficiencies are applied as correction factors, shown as horizontal markers in Fig. 3. For DDLG system, $\varepsilon_{m \rightarrow e} = 0.70$ is applied, consistent with reported efficiencies for permanent magnet linear generators operating at low velocities, according to Ahamed et al. (2022). For EGEC, $\varepsilon_{m \rightarrow e} = 0.55$ is used, reported in Li et al. (2020), which accounts for

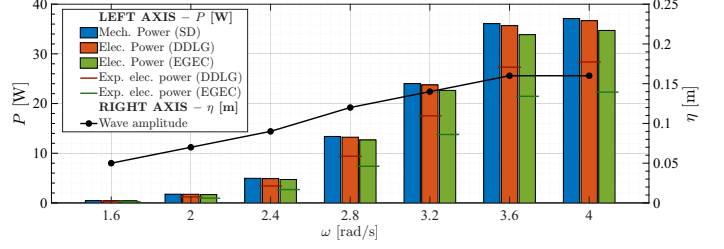


Fig. 3. Absolute mean mechanical and electrical power output P [W] as a function of wave frequency ω [rad/s] for the three PTO configurations namely Spring-Damper (SD), Direct Drive Linear Generator (DDLG) and Electric Generator Equivalent Circuit (EGEC). Blue bars represent the mechanical power of the SD configuration; orange and green bars correspond to the electrical power output of the DDLG and EGEC, respectively. Horizontal markers indicate the expected electrical power estimated from the mechanical output using a mechanical-to-electrical efficiency.

the additional losses introduced by the rack-and-pinion transmission stage. These efficiency values provide a more conservative and experimentally consistent performance baseline.

Finally, Fig. 4 shows the instantaneous power time series $P(t)$ for the wave condition $\omega = 4.0$ [rad/s] and $\eta = 0.16$ [m] over the steady-state interval $t \in [60, 70]$ [s]. All three signals exhibit periodic oscillatory behavior consistent with the sinusoidal wave excitation, with a period of approximately $T = 1.57$ [s]. The mechanical power from the SD system (blue) reaches peak amplitudes of approximately ± 40 [W], while the electrical outputs of the DDLG and EGEC systems are reduced due to their respective conversion efficiencies.

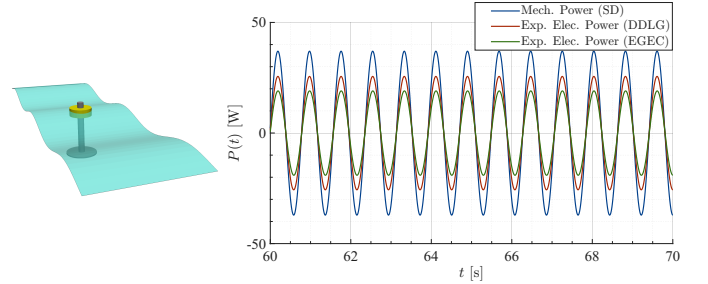


Fig. 4. Time-domain power output for the wave of $\omega = 4.0$ [rad/s] over the steady-state window $t \in [60, 80]$ [s]. The blue curve is the subtracted mechanical power absorbed by the Spring-Damper (SD) PTO, while the red and green curves are the expected electrical power output of the Direct Drive Linear Generator (DDLG) and Electric Generator Equivalent Circuit (EGEC) systems, respectively, after applying mechanical-to-electrical conversion efficiencies.

The results show that both the DDLG and EGEC architectures are capable of reproducing the target damping coefficient through adjustment of their respective electrical load resistances, confirming that the linear dampers derived in Eqs. 14 and 15 are practical and effective parameterization strategies. The DDLG exhibits a higher mechanical-to-electrical conversion efficiency than the EGEC across all wave conditions, owing to its direct coupling architecture

and the absence of an intermediate transmission stage, according to the literature, see Ahamed et al. (2022). The EGEC, while introducing additional losses through the rack-and-pinion mechanism, Li et al. (2020), but offering the advantage of two independent tuning parameters — load resistance R_{load} and pinion radius r . These characteristics suggest that the choice between architectures involves a trade-off between conversion efficiency and tunability that depends on the intended deployment scenario.

6. CONCLUSIONS AND FUTURE WORKS

This work developed and compared three PTO architectures within the time-domain WEC-Sim framework for a tunable laboratory-scale two-body PA WEC. The results confirm that WEC-Sim, combined with BEM hydrodynamic data from CAPYTAIN, provides a flexible and computationally efficient platform for PTO performance evaluation. Yet, the lumped-parameter electrical models available in PTO-Sim do not capture internal generator losses, requiring the application of efficiency corrections to produce physically representative power estimates. The upcoming experimental tests at the UNH wave tank will provide measured electrical output data for both PTO architectures, enabling a direct validation of the efficiency assumptions adopted here and a more complete characterization of the device under realistic operating conditions.

DECLARATION OF GENERATIVE AI TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the author(s) used Claude (Anthropic) and Perplexity AI to assist in improving the clarity, grammar, and overall writing quality of the manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the accuracy, integrity, and content of the published article.

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